Reproducibility of LLM-based Recommender Systems Research: Worrying Observations from an Initial Analysis

Faisal Shehzad, and Dietmar Jannach
University of Klagenfurt, Austria
faisal.shehzad@aau.at, dietmar.jannach@aau.at

Motivation

- Large language models (LLMs) are shaping recommender systems research and becoming central to modern recommendation architectures.
- Their stochasticity and complexity introduce new reproducibility challenges.
- **Goal:** Identify key reproducibility variables for LLM-based recommender systems and assess their documentation in high-ranked venues.

Research Methodology

Paper Selection Process:

- Scanned 2023–2025 proceedings of high-ranked conferences and selected papers mentioning “LLM” with “recommender” or “recommendation” in the title, abstract, or introduction.
- Excluded papers for which all authors were affiliated with industry, as code sharing is often restricted in such cases.
- This criterion leads us to the selection of 64 papers.

Table 1: Papers by venue

Venue	RecSys	SIGIR	WWW	WSDM	KDD
No. of papers	25	17	8	7	7

Reproducibility Variables

We use variables from prior studies and policies as reproducibility indicators.

With slight customization, we grouped them into two main variables.

- **General reproducibility variables** (source code, hyperparameter ranges, train-test splits, etc.)
- **LLM-specific reproducibility variables** (model version, fine-tuning materials, Fine-tuning strategy, Checkpoints, Prompt type, Prompt documentation, Temperature, etc.)

Results (1/2): General reproducibility variables

Source code

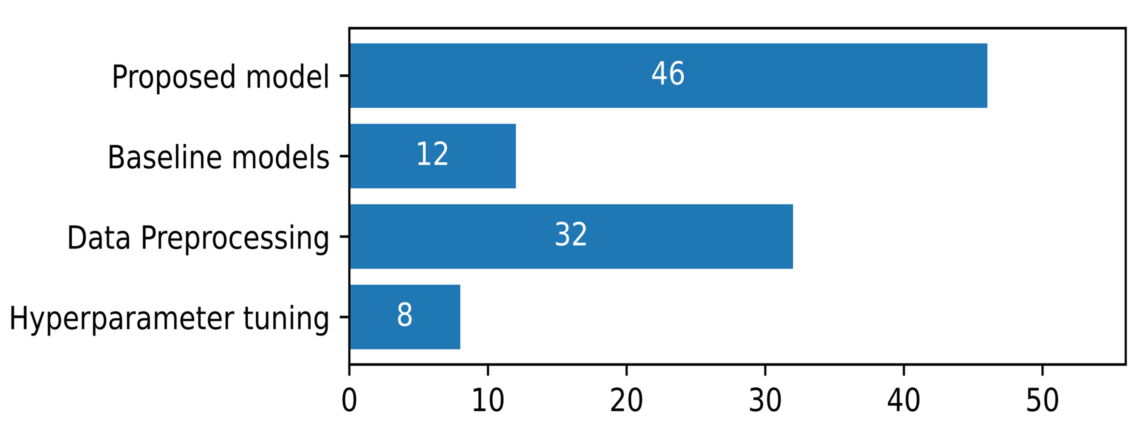

Figure 1: Code sharing statistics

Additional observations:

- **Of 64 papers**, **49** provided code repository links; however, **3** were inaccessible, leaving **46 papers (71%)** with available code.
- All repositories share proposed-model code; only **12** include baselines.
- Hyperparameter tuning was required for **59 papers**, but only **5** shared complete tuning scripts; **3** shared partial scripts, and **11** used fixed hyperparameters without any justification.
- **For the remaining reproducibility variables**, such as hyperparameter ranges, train-test splits, and evaluation details, our findings align with previous studies; therefore, they are not reported in the poster.

Results (2/2): LLM-specific reproducibility variables

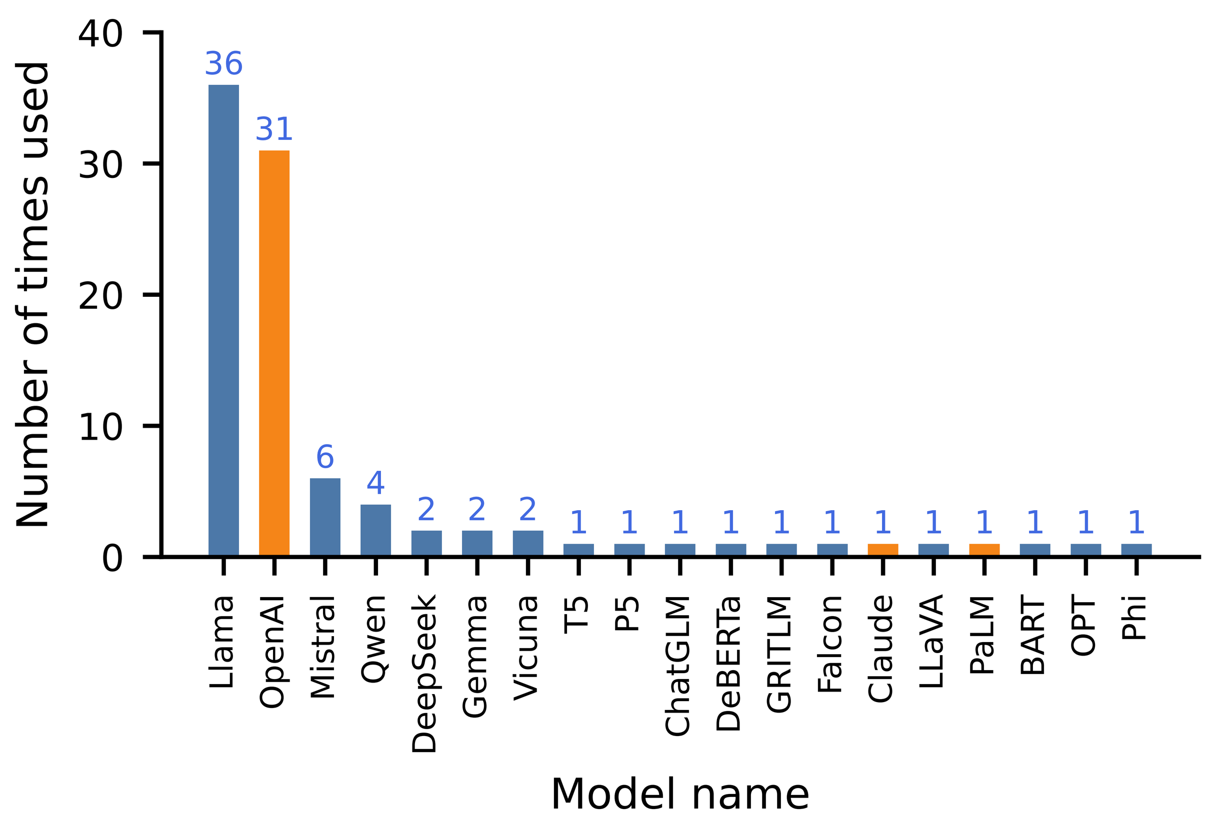

Figure 2: Frequency of model usage for closed- and open-source LLMs.

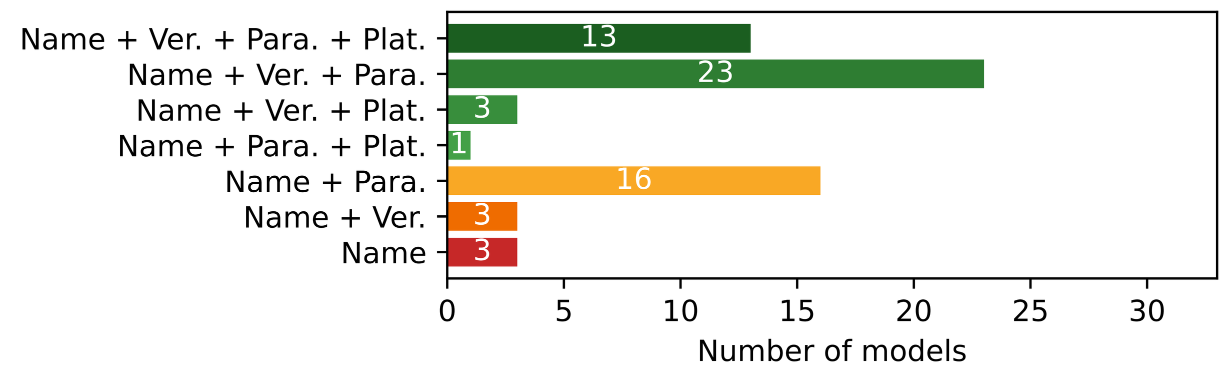

Figure 3: Documentation of Open-Source Model Usages.

LLM Models and Versions:

- Most studies **(38 papers)** use open-source models; **13** use closed-source models, **12** use both, and **1** does not report the model name.
- OpenAI models are used **31 times across 24 papers**, but only **5 papers** report both version and snapshot; **24** report only the version, and **1** reports only the model name.
- Open-source models are used **62 times in 52 papers**, with **Llama** dominating; Figure 3 summarizes how well key details are reported.

Fine-Tuning Aspects:

- Fine-tuning is used in **39 papers**, but only **6 (15%)** provide complete tuning materials; the rest provide partial or no materials.
- Checkpoints and model weights support reproduction, but only **1 paper** shares checkpoints and **3** share model weights.
- Tokenizer details are rarely reported: only **6 papers** provide both tokenizer name and version.

Prompt-Related Aspects:

- Prompts are used in **50 papers**; **47** document them, **1** partially, and **2** provide none.
- Prompt templates are shared in **44 papers**; among **34 papers** with source code, **more than half (23)** include the exact template in the code.
- Only **5** papers report the context window length for prompt templates.

Temperature affects LLM output randomness and reproducibility; **only 43.75% of papers** report its value.

Conclusion & Discussion

- Our findings align with prior evidence: current experimental reporting does not fully support reproducibility.
- Long-standing reproducibility issues persist in LLM-based studies and are further compounded by LLM-specific documentation needs.
- Stricter and more rigorous practices are needed to improve reproducibility.